



MATHEMATICS METHODS : UNITS 1 & 2, 2023

Test 2 – (10%)

1.2.5, 1.2.9 to 1.2.18, 1.2.22 to 1.2.23, 1.3.7 to 1.3.16,
1.1.4 to 1.1.11

PM

Time Allowed 20 minutes	First Name SOLUTIONS	Surname	Marks 22 marks
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Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
Mr Lockyer Mr Luzuk Mr Mohammadi
Ms Narendranathan Mr Ng
Mr Stephens Mr Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 [4 marks]

A cubic polynomial $f(x)$ satisfies the following conditions:

- $f(-4) = 0$,
- $f(0) = -2$ and
- it has a repeated root at $x = 1$.

Find the equation of $f(x)$ in the form $ax^3 + bx^2 + cx + d$.

$$f(x) = a(x + 4)(x - 1)^2$$

$$\text{Substitute } (0, -2) \text{ into } f(x): -2 = a(4)(-1)^2$$

$$\Rightarrow a = -\frac{1}{2}$$

$$\therefore f(x) = -\frac{1}{2}(x + 4)(x - 1)^2$$

$$= -\frac{1}{2}(x + 4)(x^2 - 2x + 1)$$

$$= -\frac{1}{2}(x^3 + 2x^2 - 7x + 4)$$

$$= -\frac{1}{2}x^3 - x^2 + \frac{7}{2}x - 2$$

Question 2 [9 marks: 2, 4, 3]

Solve the following equations:

a) $3(5 - x)(3x + 4) = 0$

$$x = 5 \text{ or } -\frac{4}{3}$$

b) $x = 2x^3 + 6x^2$

$$\Rightarrow x(2x^2 + 6x - 1) = 0$$

$$\Rightarrow x = 0 \text{ or } 2x^2 + 6x - 1 = 0$$

$$\begin{aligned} \text{Use Quadratic formula: } x &= \frac{-6 \pm \sqrt{6^2 - 4 \times 2 \times (-1)}}{2 \times 2} \\ &= \frac{-6 \pm 2\sqrt{11}}{2 \times 2} \\ &= \frac{-3 \pm \sqrt{11}}{2} \text{ or } \frac{-3 - \sqrt{11}}{2} \end{aligned}$$

c) $\cos 2x = -\frac{\sqrt{2}}{2}$ for $0 \leq x \leq \pi$.

$$0 \leq 2x \leq 2\pi$$

$$\text{Reference } \angle = \frac{\pi}{4}$$

$$2x = \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}$$

$$x = \frac{3\pi}{8}, \frac{5\pi}{8}$$

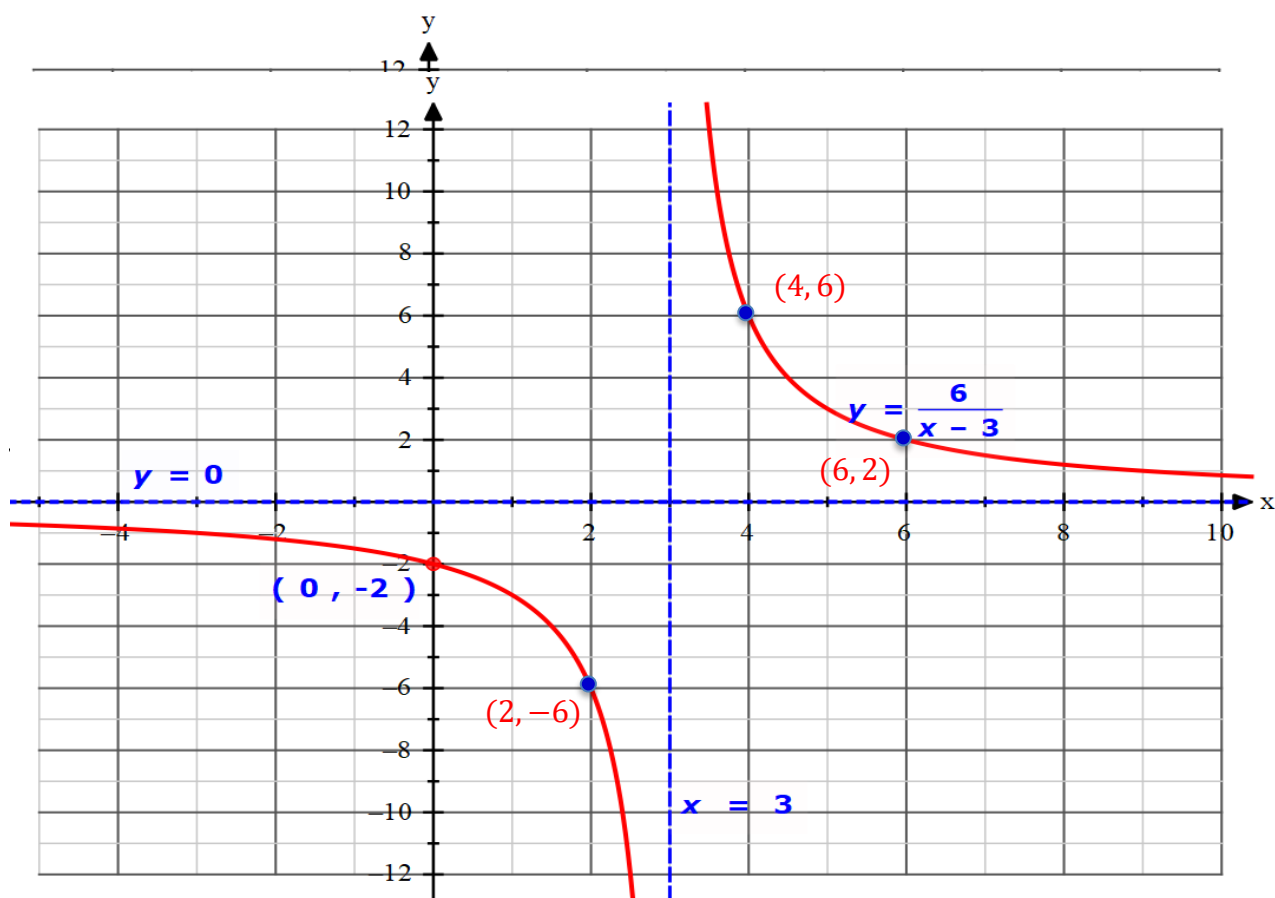
Question 3 [4 marks: 1, 3]

Given that $f(x) = \frac{6}{x-3}$.

a) State the equations of the asymptotes.

$x = 3 \text{ and } y = 0$

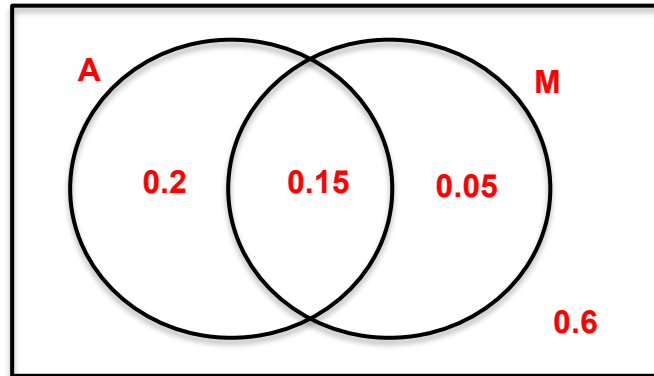
b) Sketch the graph of $f(x)$ on the axes provided, clearly labelling the key features.



Question 4 [5 marks: 2, 1, 2]

Jon has completed his Semester 1 exams for Mathematics Methods and Applications. The probability of him getting a 'B' grade in Methods is 0.2 and in Applications is 0.35. There is a 60% chance that he will not get a 'B' in either course.

- a) Represent the above information on the Venn diagram below by clearly labelling the events and placing probabilities in all regions.



- b) What is the probability that Jon will get a 'B' in both courses?

$$P(A \cap M) = 0.15$$

- c) What is the probability that he gets a 'B' for Methods given that he only got one 'B'?

$$P(B \text{ for Methods given } B \text{ for only one}) = \frac{0.05}{0.2+0.05} = \frac{1}{5} \text{ (or 0.2)}$$

End of Calculator Free Section



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1.1.4 to 1.1.11

PM

Time Allowed	First Name	Surname	Marks
30 minutes			30 marks

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- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

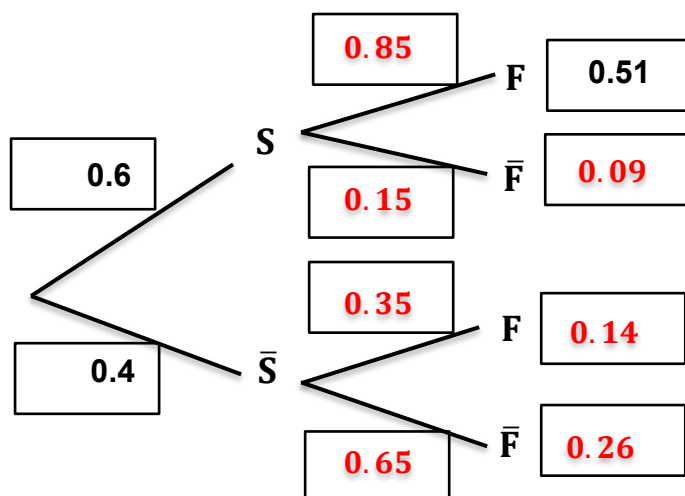
PART B – CALCULATOR ASSUMED

All approximate answers to be stated to an accuracy of 2 decimal places, unless otherwise directed.

Question 5 [7 marks: 3,4]

A survey about Year 12 leavers travel plan showed that 60% had an overseas destination (event S) and that 65% had travelled with friends (event F). Also, 51% of the leavers had travelled overseas with friends.

a) Use the above information to complete all the probabilities in the tree diagram.



b) Determine the probability that a randomly selected Year 12 leaver:

(i) Travelled with friends given that they did not travel overseas.

0.35

(ii) Travelled with friends or overseas.

$0.6 + 0.14 = 0.74$

(iii) Did not travel overseas given that they travelled with friends, leaving your answer as a simplified fraction or correct to 4 decimal places.

$\frac{0.14}{0.65} = \frac{14}{65} \approx 0.2154 \text{ (to 4 d.p.)}$

Question 6 [4 marks]

If $\sin X = -\frac{\sqrt{7}}{4}$ and $\cos Y = \frac{21}{29}$ with X and Y in the same quadrant, find, leaving your answer in exact value, the value of $\cos(X + Y)$.

$$\sin X = -\frac{\sqrt{7}}{4} \Rightarrow \cos X = \frac{\sqrt{4^2 - 7}}{4} = \frac{3}{4}$$

$$\cos Y = \frac{21}{29} \Rightarrow \sin Y = -\frac{\sqrt{29^2 - 21^2}}{29} = -\frac{20}{29}$$

$$\cos(X + Y) = \cos X \cos Y - \sin X \sin Y$$

$$= \frac{3}{4} \times \frac{21}{29} - \left(-\frac{\sqrt{7}}{4}\right) \left(-\frac{20}{29}\right)$$

$$= \frac{63}{116} - \frac{5\sqrt{7}}{29}$$

Question 7 [4 marks: 2, 2]

The equation of a circle is given by $x^2 + y^2 + 2x - 4y = 4$.

a) Find the centre and radius of this circle.

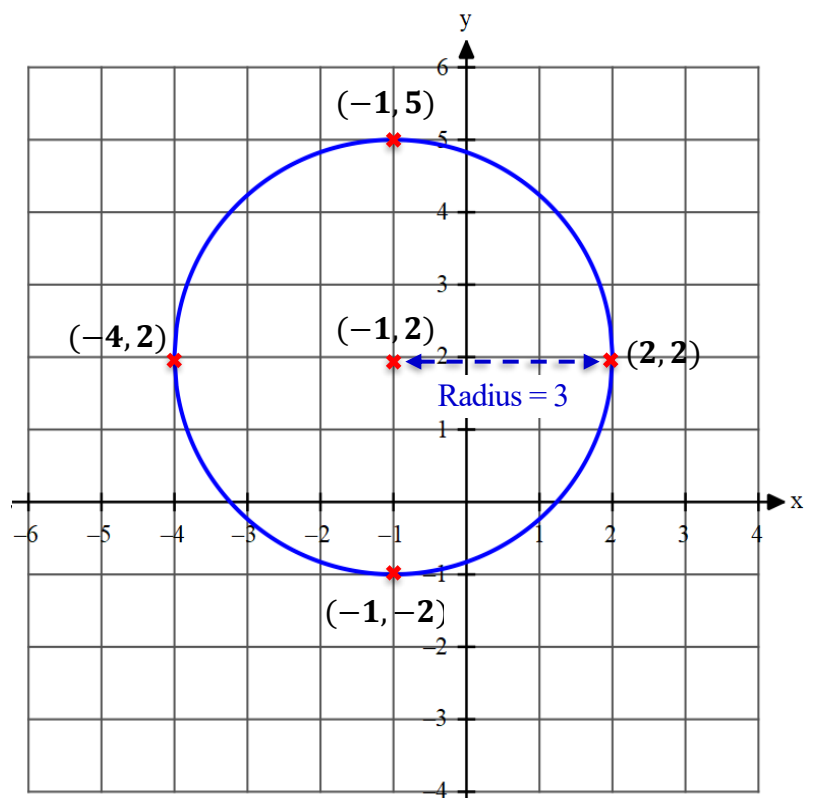
$$x^2 + y^2 + 2x - 4y = 4$$

$$\Rightarrow (x + 1)^2 - 1 + (y - 2)^2 - 4 = 4$$

$$\Rightarrow (x + 1)^2 + (y - 2)^2 = 3^2$$

Centre $(-1, 2)$, radius = 3

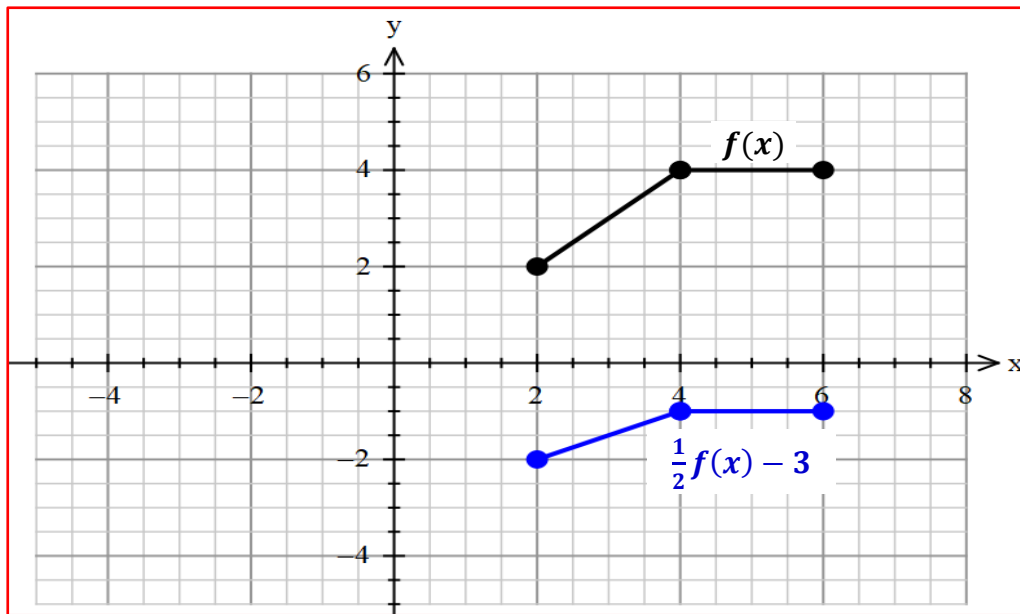
b) Sketch the circle on the grid given, indicating the centre and radius.



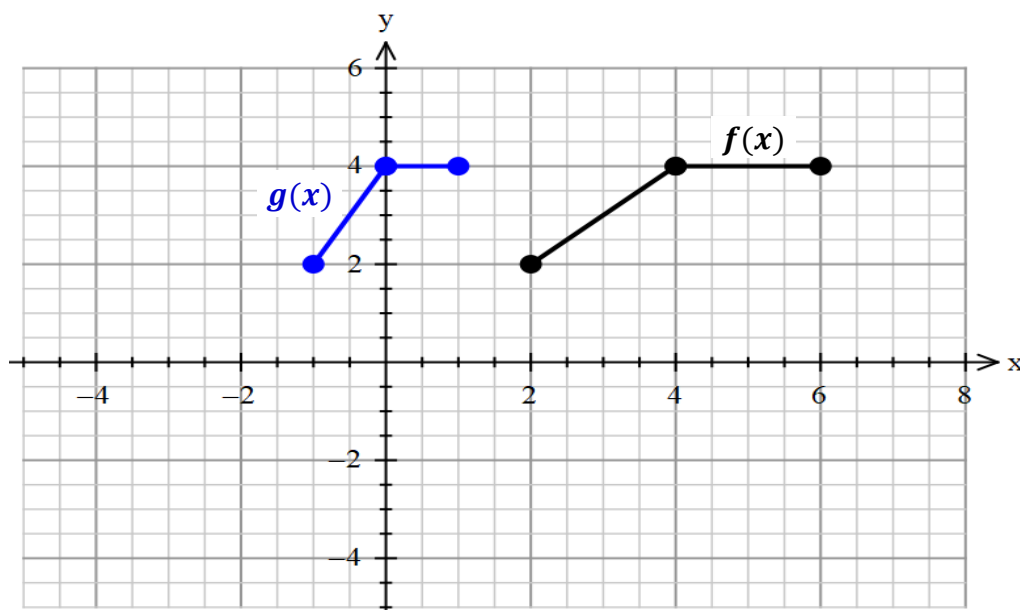
Question 8 [6 marks: 2,2,2]

The graph of the function $y = f(x)$ is shown below.

- a) Draw, on the same grid, the graph of $y = \frac{1}{2}f(x) - 3$.



- b) $f(x)$ and $g(x)$ are graphed below.



- (i) State the transformations, in order, that map $f(x)$ to $g(x)$.

1. Horizontal translation: 4 units left
2. Horizontal dilation: scale factor $\frac{1}{2}$ (or 0.5)

OR

1. Horizontal dilation: scale factor $\frac{1}{2}$ (or 0.5)
2. Horizontal translation: 2 units left

- (ii) State $g(x)$ as a function of $f(x)$.

$g(x) = f(2x + 4)$ OR
 $g(x) = f(2(x + 2))$

Question 9 [9 marks: 1,3,1,2,2]

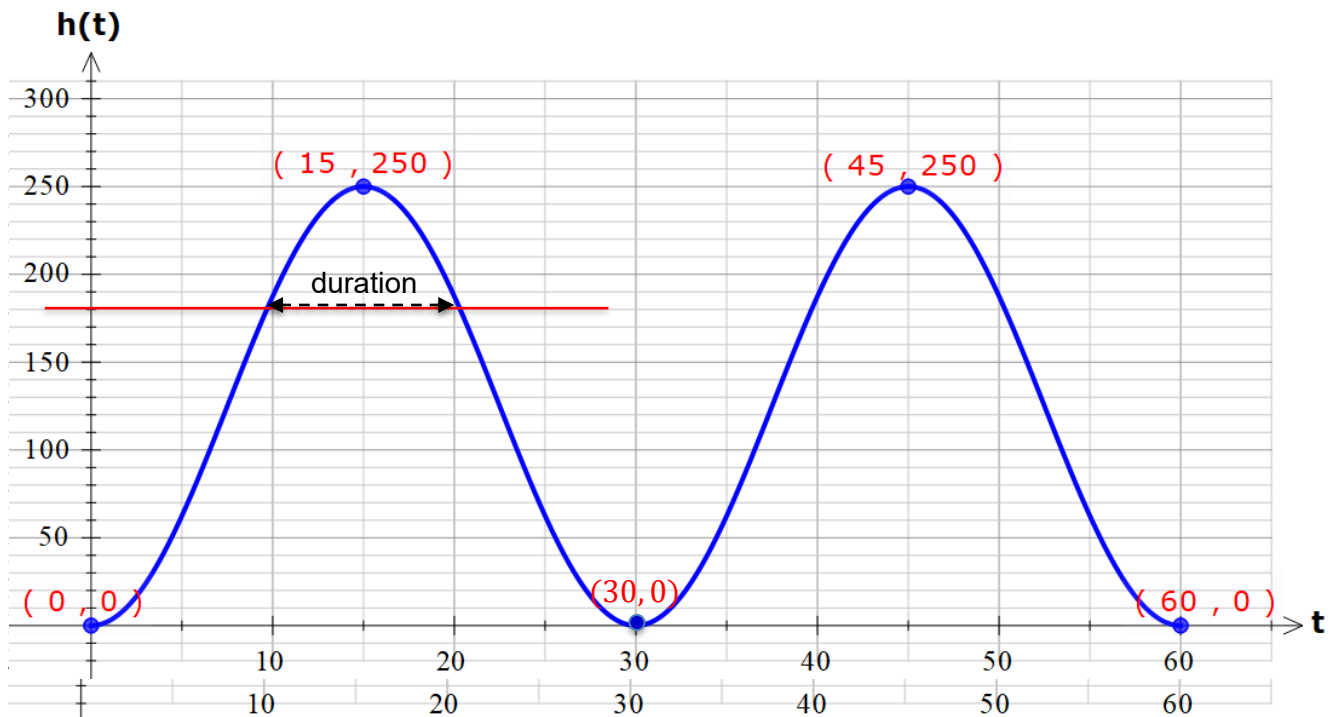
The Dubai Eye is the largest Ferris wheel in the world, and it moves in a vertical circle. To provide a different experience for the passengers, the function of motion has been re-configured to

$h(t) = 125 \left(1 - \cos \left(\frac{\pi}{15} t \right) \right)$, where h measures the height above the ground in metres and t is the time in minutes since the measurements began.

- a) Determine the maximum height that can be reached by the Dubai Eye.

250 m

- b) Sketch the graph of $h(t)$ on the axes below for $0 \leq t \leq 60$.



- c) What is the period of the motion?

30 minutes

- d) Starting from the ground level, Jerry had an hour's ride on the Dubai eye. Determine the time(s) where he is at the highest point.

15 and 45 minutes

- e) Tom completed one revolution of the ride where he also started at the ground level. What was the total duration that he was at a height of at least 180 metres above the ground during his ride?

Solve $125 \left(1 - \cos \left(\frac{\pi}{15} t \right) \right) = 180$ for $0 \leq t \leq 30 \Rightarrow t = 9.6753$ or 20.3247
Duration $\approx 20.3247 - 9.6753 \approx 10.65$ minutes (to 2 d.p.)

End of Calculator Assumed Section